Waste Sorting — experiment report

This report is generated from `runs/<run>/val_metrics.json`, `test_metrics.json`, `train_log.json`. Validation metrics are used for model selection; test metrics are for the final held-out comparison.

Test-set comparison

config	backbone	strategy	seed	val_macro_f1	test_accuracy	test_macro_f1	latency_ms_
clip_vitb32_full_finetune	clip_vit_b32	full_finetune	87578	0.9568	0.9499	0.9500	
convnext_tiny_full_finetune	convnext_tiny	full_finetune	87578	0.9654	0.9499	0.9428	
convnext_tiny_layer_decay	convnext_tiny	layer_decay_finetune	87578	0.9669	0.9314	0.9330	
convnext_tiny_gradual_unfreeze	convnext_tiny	gradual_unfreeze	87578	0.9568	0.9420	0.9316	
effb0_finetune_seed6545656				nan	0.9182	0.9146	
effb0_finetune_seed87578				nan	0.9182	0.9144	
effb0_finetune_seed213123				nan	0.9156	0.9086	
resnet50_finetune_seed87578				nan	0.9129	0.9080	
resnet50_finetune_seed213123				nan	0.9288	0.9074	
effb0_full_finetune	efficientnet_b0	full_finetune	87578	0.9507	0.9129	0.9060	
resnet50_full_finetune	resnet50	full_finetune	87578	0.9239	0.9208	0.9060	
resnet50_finetune_seed6545656				nan	0.9050	0.8805	
clip_vitb32_feature_extraction	clip_vit_b32	feature_extraction	87578	0.8949	0.8813	0.8717	
effb0_feature_extraction_seed6545656				nan	0.8707	0.8471	
effb0_feature_extraction_seed213123				nan	0.8707	0.8467	
effb0_feature_extraction	efficientnet_b0	feature_extraction	87578	0.8396	0.8549	0.8310	
resnet50_feature_extraction	resnet50	feature_extraction	87578	0.8504	0.8628	0.8300	
resnet50_feature_extraction_seed6545656				nan	0.8575	0.8259	
resnet50_feature_extraction_seed213123				nan	0.8575	0.8254	

Highest test macro-F1 (descriptive)

- Run: `clip_vitb32_full_finetune` (seed `87578`)
- Backbone / strategy: `clip_vit_b32` / `full_finetune`
- Test accuracy: 0.9499
- Test macro-F1: 0.9500

Files

- `comparison.csv` — complete tabular metrics, including per-class precision/recall/F1 and TTA runs.
- Each run folder — confusion matrices, misclassified examples, model checkpoints, training history and metrics JSON.